

555 Timer IC

Precision Timer / PWM Generator

Part Number: NE555 (Texas Instruments)

1. Component Name

NE555 Precision Timer (family also includes NA555, SA555, SE555), a general-purpose, industry-standard 8-pin timer IC.

2. Introduction

The 555 Timer is one of the most widely used analog/digital hybrid integrated circuits in electronics. It can generate accurate time delays or oscillations and is commonly configured in astable mode to produce a continuous square-wave/PWM-like output, in monostable mode to produce a single timed pulse, or as part of pulse-width-modulation and pulse-position-modulation circuits.

3. Manufacturer

Manufacturer	Texas Instruments
Official Datasheet	NA555, NE555, SA555, SE555 Precision Timers
Datasheet Reference	ti.com/lit/ds/symlink/ne555.pdf

4. Functional Description

Working Principle

Internally the 555 consists of two comparators, an SR flip-flop, and a discharge transistor. In astable operation the trigger level is approximately one-third of the supply voltage and the threshold level is approximately two-thirds of the supply voltage. A timing capacitor charges through external resistors until it reaches the threshold voltage, at which point the flip-flop resets the output low and the discharge pin conducts, allowing the capacitor to discharge until the trigger level is reached again, restarting the cycle. This produces a continuous rectangular output waveform whose frequency and duty cycle are set by the external resistor-capacitor network.

Purpose in PWM Generation

In astable mode, a 555 timer is a simple, microcontroller-free way to generate a fixed-frequency PWM signal for motor speed control: the duty cycle set by the external resistor-capacitor network determines the average voltage delivered to the load. This makes it a common building block in H-Bridge DC motor driver circuits that do not use a programmable microcontroller for PWM generation.

5. Key Features

Feature	Specification
Operating Voltage	4.5 V to 16 V (SE555: up to 18 V)
Supply Current	Approx. 2 mA (typical, low supply)
Output Current	Up to 200 mA (sink or source)

Feature	Specification
Frequency Range	Sub-1 mHz up to approximately 100 kHz (application dependent)
Switching Characteristics	Output compatible with TTL logic levels at 5 V supply
Package Types	PDIP-8, SOIC-8 and other 8-pin variants
Operating Temperature	0 degC to 70 degC (commercial NE555); wider ranges for SA555/SE555
Timing Accuracy	Independent of supply voltage once stable (RC-determined)

6. Pin Configuration (8-pin DIP/SOIC package)

Pin Number	Pin Name	Function
1	GND	Ground reference
2	TRIG	Trigger input; output set high when below 1/3 VCC
3	OUT	Output pin; sources/sinks up to 200 mA
4	RESET	Active-low reset; overrides all other inputs
5	CONT	Control voltage; allows external modulation of threshold levels
6	THRES	Threshold input; output reset low when above 2/3 VCC
7	DISCH	Discharge; open-collector path to ground when output is low
8	VCC	Positive supply voltage

7. Electrical Specifications

Parameter	Symbol	Value	Conditions
Supply Voltage Range	VCC	4.5 V - 16 V	SE555 max 18V
Output Current (max)	IOUT	200 mA	Sink or source
Supply Current	ICC	~2 mA (typ.)	Low supply condition
Threshold/Trigger Ratio	-	0.67 VCC / 0.33 VCC	Astable/monostable operation
Operating Temperature (NE555)	TA	0 degC to 70 degC	Commercial grade
Timing Formula (Astable)	f	$f = 1.44 / [(RA+2RB)C]$	Standard astable configuration

8. Typical Applications

- Astable square-wave / PWM signal generators
- Monostable pulse and timing-delay circuits
- LED flasher and blinker circuits
- Tone and frequency generators
- Missing-pulse and frequency-divider detectors
- Simple motor speed controllers (as a 555-based PWM source)
- Touch and proximity sensing circuits
- Sequential timers and pulse-position modulators

9. Advantages

- Extremely low cost and widely available
- Simple external component count (a few resistors and a capacitor)
- No firmware/programming required for basic timing functions
- Robust output stage capable of directly driving small loads (up to 200 mA)
- Well documented with decades of established reference designs

10. Limitations

- Fixed analog RC-based timing is less precise and less flexible than a microcontroller-generated PWM
- Duty cycle adjustment in the basic astable configuration is limited without additional diodes/components
- No built-in capability for closed-loop feedback or dynamic frequency changes
- Cannot directly interface with digital control logic without extra circuitry

11. Role in This Project

This project's PWM signal is generated by the STM32F103C6 microcontroller rather than a 555 timer; this datasheet is included as a reference for the 555 timer as a common alternative PWM source in H-Bridge DC motor driver designs.

12. Precautions

- If a 555 timer were to be substituted into a design, decouple VCC to GND with a 0.1 uF ceramic capacitor close to the IC
- Keep timing resistors within a sensible range (typically 1 k-ohm to 1 M-ohm) for stable operation
- Tie unused control inputs to an appropriate logic level to prevent false triggering
- Observe the maximum output current rating (200 mA) to avoid damaging the output stage
- Respect the recommended operating temperature range for the selected grade (NE/SA/SE)

13. Official Reference

Official Manufacturer	Texas Instruments
Official Datasheet Title	NA555, NE555, SA555, SE555 Precision Timers
Datasheet Version	Rev. K (March 2026), Texas Instruments (SLFS022)